

MEDICINE HAT, AB, Canada

WMO: 712470

Lat: 50.0250N

Lon: 110.7172W

Elev: 717

StdP: 93.00

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.8	-26.0	-34.4	0.2	-29.6	-30.1	0.3	-25.8	13.1	4.8	11.7	2.9	2.1	220	0.651

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	33.0	17.3	31.1	16.9	29.2	16.4	19.0	28.7	18.1	28.0	17.2	26.9	4.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	12.2	22.0	14.6	11.3	21.2	13.6	10.6	20.4	56.9	28.8	53.8	27.8	51.1	27.1	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.7	8.5	-33.7	36.4	3.0	1.4	-35.8	37.4	-37.5	38.2	-39.2	39.0	-41.4	40.0		
(5)			-33.8	21.1	2.9	1.0	-35.9	21.8	-37.6	22.4	-39.3	23.0	-41.4	23.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-8.4	-6.7	-0.5	6.4	12.0	16.3	20.4	19.4	14.0	6.7	-1.0	-7.1		
	DBStd	11.90	9.18	8.46	7.94	5.04	4.56	3.46	3.27	3.64	4.74	5.19	7.33	8.39		
	HDD10.0	2532	572	468	327	127	31	1	0	0	16	126	334	532		
	HDD18.3	4676	830	701	583	358	201	79	16	31	144	361	581	790		
	CDD10.0	1079	0	0	2	19	93	191	321	292	136	23	3	0		
	CDD18.3	182	0	0	0	0	5	19	79	65	14	1	0	0		
	CDH23.3	2748	0	0	1	11	121	292	1091	943	279	11	0	0		
	CDH26.7	1043	0	0	0	1	26	79	447	392	97	1	0	0		
(14) Wind	WSAvg	3.8	3.8	3.6	4.0	4.2	4.0	3.8	3.5	3.3	3.5	4.0	4.0	3.9		
(15) Precipitation	PrecAvg	347	21	12	17	29	44	57	42	33	38	17	16	19		
	PrecMax	554	61	44	44	97	91	132	90	94	202	39	41	54		
	PrecMin	222	2	0	1	1	8	9	8	6	3	1	2	4		
	PrecStd	81	16	9	11	21	22	35	24	22	39	11	11	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	12.6	19.0	24.8	29.7	32.2	35.4	35.3	32.8	24.8	17.9	11.0		
		MCWB	5.2	5.9	9.0	11.7	15.1	17.4	18.3	17.6	16.1	12.8	9.0	5.1		
	2%	DB	7.2	8.5	15.6	21.4	26.6	28.9	33.2	32.9	29.5	21.1	14.1	8.2		
		MCWB	3.3	3.5	7.5	10.5	14.0	16.5	18.1	16.8	15.1	11.1	7.1	3.6		
	5%	DB	5.3	6.0	12.7	18.8	24.0	26.6	31.2	31.0	26.7	18.1	11.2	5.9		
		MCWB	2.2	2.1	5.9	9.1	12.8	15.6	17.6	16.6	14.3	10.0	5.7	2.3		
	10%	DB	3.7	3.9	9.9	15.9	21.4	24.5	29.2	28.7	23.7	15.4	8.5	4.0		
		MCWB	1.3	1.1	4.4	7.5	11.5	14.7	17.1	16.1	13.0	8.5	3.9	0.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	6.3	9.7	12.5	16.1	19.4	20.9	19.3	17.4	13.7	9.9	5.9		
		MCDB	9.5	11.5	18.2	23.3	26.9	28.2	31.0	29.6	28.8	23.1	16.5	10.0		
	2%	WB	3.6	3.8	7.9	11.0	14.6	17.7	19.5	18.2	15.9	11.8	7.6	3.9		
		MCDB	6.8	8.0	14.7	20.7	24.4	26.1	29.1	28.8	26.9	19.6	13.6	7.6		
	5%	WB	2.4	2.2	6.2	9.4	13.4	16.5	18.6	17.5	14.7	10.2	5.9	2.4		
		MCDB	5.0	5.5	12.1	17.7	22.4	24.5	28.2	28.0	25.2	17.2	10.7	5.8		
	10%	WB	1.3	1.1	4.5	8.0	12.3	15.3	17.7	16.7	13.5	8.9	4.2	0.9		
		MCDB	3.8	3.9	9.5	15.0	20.1	23.0	27.3	26.8	22.9	14.9	8.3	3.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	10.7	11.5	12.9	13.1	12.6	14.7	14.9	14.1	12.7	10.8	10.2		
		MCDBR	10.6	12.7	15.8	18.0	17.6	16.3	17.7	18.9	19.6	17.0	14.2	11.1		
		MCWBR	7.8	8.8	9.2	9.3	7.8	6.9	6.6	7.0	8.4	9.0	8.6	7.7		
		MCWBR	10.5	12.1	15.1	16.7	15.5	14.6	15.6	16.2	18.1	16.2	13.5	11.1		
(40) Clear-Sky Solar Irradiance	taub	taub	0.271	0.267	0.301	0.359	0.383	0.373	0.372	0.394	0.344	0.303	0.270	0.257		
		taud	2.389	2.469	2.439	2.275	2.252	2.344	2.361	2.267	2.388	2.485	2.479	2.408		
		Ebn at Noon	785	895	914	884	873	885	879	837	850	835	789	753		
		Edh at Noon	62	74	92	121	130	120	116	121	96	73	57	53		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.23	2.22	3.57	4.94	5.86	6.37	6.93	5.59	3.99	2.47	1.38	0.96		
		RadStd	0.12	0.19	0.25	0.34	0.41	0.38	0.34	0.35	0.33	0.25	0.10	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page